

Python Interview Preparation

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1. What is python ?

Answer : Python ek high-level, interpreted, object-oriented programming language hai. Ye simple syntax ki wajah se famous hai aur web development, AI, automation, scripting aur data science me use hoti hai.

Features:

- Easy syntax
- Readable code
- Large community support
- Cross-platform
- Huge libraries support

Example:

```
print("Hello World")
```

2. Python Interpreted Language Kyu Kehlati Hai?

Answer: Python interpreted language hai kyun ke iska code line by line execute hota hai compiler ki tarah pehle pure program ko machine code me convert nahi karta.

3. What are Data Types in Python?

Answer: Python me different types ka data store karne ke liye built-in data types hote hain. Inhein different categories me divide kiya jata hai.

Types of Data Types in Python

Category	Data Types	Example
Numeric	int, float, complex	10, 4.12, 2+3j
Sequence	list, tuple, range	[1,2], (1,2)
Text	String	"Hello"
Set	Set	{1,2,3}
Mapping	Dict	{"a" : 1}
Boolean	bool	True, False
Binary	bytes, bytearray, memoryview	b"hello"

Python may built-in data types wo hote hain jo different types ka data store krte hain.

4. Variables kia hote hain?

Answer: variable memory location ka nam hote hain jis may data store hota hai

5. What is mutable and immutable in python?

Answer:

i. mutable wo data types hote hain jin ki value ko change kia ja skta hai

Example:

- List
- Dict
- Set

ii. immutable data types wo hote hain jin ki value ko change nahi kia ja skta

Example:

- Typle
- String
- Int

6. What is the difference between list and tuple?

Answer:

i. List mutable hai isko square brackets may define kia jata hai or ye tuple se thori slow hoti hain

ii. tuple immutable hai or isko parentheses may define kia jata hai tuple list se thori fast hoti hai

7. What is type casting in python?

Answer: aik data type ko dusre data type may convert krna type casting kehlata hai python may do tarha ki type casting hoti hai

i. Implicit

Python automatically convert krta hai jb zarorat ho

ii. Explicit

Hum manually conversion krte hain

8. Operators in python ?

Answer:

i. Arithmetic operators

+, -, *, /, //, %, **

ii. Comparison Operators

==, !=, >, <, >=, <=

iii. Logical Operators

And, or, not

9. What is a function?

Answer: function reusable block of code hota hai jo specific task ko perform krta hai

Example:

```
def greet(name):  
    return f"hello{name}"
```

10. What is difference between parameter and arguments:

Answer: **Parameters** function k andar define hote hain (Variables)

Arguments function ko call krte hain waqt pass kiy jate hain (values)

Types of Arguments

- Positional Arguments
- Keyword arguments
- Default arguments
- variable-length Arguments

Examples:

i. Positional Arguments

Defination: jaha order tarteeb matter krta ho

```
def student(name, age, city):  
    print(f"{name} is {age} years old from {city}")
```

Sahi - Order match kar raha hai

```
student("Ali", 20, "Karachi")
```

Output: Ali is 20 years old from Karachi

Ghalat - Order mismatch

```
student("Karachi", "Ali", 20)
```

Output: Karachi is Ali years old from 20 (meaningless)

ii. Keyword Arguments

Defination: jaha parameter name explicitly likhte hain

```
def student(name, age, city):  
    print(f"{name} is {age} years old from {city}")
```

Order koi bhi ho sakta hai

```
student(age=20, city="Lahore", name="Sara")
```

Output: Sara is 20 years old from Lahore

Mix bhi kar sakte hain (positional pehle, keyword baad mein)

```
student("Ahmed", city="Islamabad", age=22)
```

Output: Ahmed is 22 years old from Islamabad

iii. Default Arguments

Defination: jaha parameter ki pehle se value set ho

```
def greet(name, message="Hello"):  
    print(f"{message}, {name}!")
```

Bina message diye - default use hoga

```
greet("Ali")
```

Output: Hello, Ali!

Message diya - custom use hoga

```
greet("Sara", "Assalam Alaikum")
```

Output: Assalam Alaikum, Sara!

Multiple default arguments

```
def student(name, age=18, city="Karachi"):
    print(f"{name} is {age} years old from {city}")

student("Ahmed")           # age=18, city=Karachi
student("Ahmed", 20)       # city=Karachi
student("Ahmed", 20, "Lahore") # sab custom
```

Iv. Variable-Length Arguments

*A) args (Positional Variable-length)

Definition: jahan jaha multiple values bina key k pass krni ho

```
def sum_all(*args):
    print(f"Numbers received: {args}") # args is a TUPLE
    total = 0
    for num in args:
        total += num
    return total

# 2 arguments
print(sum_all(5, 10))
# Output: 15

# 5 arguments
print(sum_all(1, 2, 3, 4, 5))
# Output: 15

# 0 arguments bhi de sakte hain
print(sum_all())
# Output: 0
```

**B) kwargs (keyword variable-length)

Definition: jahan multiple key-value pairs pass krni ho

```
def print_details(**kwargs):
    print(f"Data received: {kwargs}") # kwargs is a DICTIONARY
    for key, value in kwargs.items():
        print(f"{key} = {value}")

# Multiple keyword arguments
print_details(name="Ali", age=20, city="Karachi", profession="Developer")
# Output:
# name = Ali
# age = 20
# city = Karachi
# profession = Developer

# 0 arguments bhi de sakte hain
```

```
print_details()
# Output: (kuch print nahi hoga)
```

11. What is the lambda function?

Answer : lambda aik anonymous single-line function hota hai

Examples:

```
square = lambda x: x **2
print(square(5)) # output 25
```

12 what is *args and **kwarg?

Answer:

i. *args

multiple Positional arguments store karta hai

ii. **kwargs

Multiple keyword arguments receive karta hai

13. What is scope?

Answer: Scope variable ki accessibility define karta hai. Python may main do scope hote hain

i. Local Scope variable jo function k andar define hota hai, siraf us function k andar accessible hota hai

ii. Global Scope variable jo function k bahir define hota hai, puri file may accessible hota hai

Important point: agr function k andar global variable ko modify krna hoto 'global' keyword use hota hai warna python usay local variable assume kr leta hai.

Examples:

Predict the output:

```
x = 100      # Global

def func1():
    x = 50    # Local (different from global x)
    print(x)  # ?

def func2():
    global x
    x = 200   # Modifying global x
    print(x)  # ?

func1()
func2()
print(x)     # ?
```

Answers: 50, 200, 200

Common interview trap question

```
my_var = 10
```

```
def test():
    print(my_var) # Line 1
    my_var = 20   # Line 2
    print(my_var) # Line 3
```

test()

Question: output kia hoga

Answer: unboundlocalError

Reason: python function k andar my_var = 20 dekh kr usay local variable treat krta hai line 1 pe print krte waqt local variable exist nahi krta is lia error ata hai.

Solution:

```
my_var = 10
```

```
def test():
    global my_var # Add this line
    print(my_var) # 10
    my_var = 20   # modifies global
    print(my_var) # 20
```

test()

14. What is recursion?

Answer: jb function khud ko call kre to recursion kehlata hai is k do parts hote hain

- Base case: (rukne ki condition)
- Recursive case (khud ko call krne ki condition)

Example:

```
def factorial(n):
    if n == 1:
        return 1
    return n * factorial(n-1)
```

15. What is exception handling

Answer: program crash hone se bachane k liye exceptions handle krte hain

Keywords:

- try
- except
- finally
- Raise

16. Difference between == and is?

Answer: i. == value compare krta hai ii. is memory location compare krta hai

Example:

```
a = [1,2]
b = [1,2]

print(a == b)
print(a is b)
```

17. Deep copy vs shallow copy

Answer: i. deep copy

Nested objects k reference ki copy banati hai agr original objects may changes hoto copy may b reflect hote hain

ii. Shallow copy

Naya object create krte hain jo original object k nested objects ki copy banati hain agr original may changes hoto copy may changes reflect nahi hote

18. Iterable vs Iterator ?

Answer: i. iterable

Iterable aik aisa object jise hum loop k zariye iterate kr skte hain

ii. Iterator

Iterator wo object hota hai jo ek ek value return karta hai using next() function.

19. What is a generator?

Answer: generator aik aisa function hai jo yield keyword use krta hai or value one by one return krta hai

20. What is a decorator?

Answer: decorator aik function hai jo kisi dusre function ki functionality ko bina change kia modify ya extend krta hai

21. What is oop?

Answer: OOP (Object-Oriented Programming) aik programming paradigm hai

Jis may real-world chezo ko objects or classes ki form may represent krte hain

22. Class vs object?

Answer: i. class blueprint hoti hai ii. Object real instance hota hai

23. What is __init__?

Answer: __init__ python may aik special method hota hai jo object create hote hee call hota hai.

24. Pillars of OOP?

Answer: i. Encapsulation ii. Inheritance iii. Polymorphism iv. Abstraction

i. Encapsulation aik oop concept hai jis may data or methods ko aik class k andar band kia jata hai or direct access se protect kia jata hai

a) Why secure data and methods?

Data or methods ko secure is lia kia jata hai take unauthorized access na ho data safe rahe

b) What is Getter and Setter?

Getter or Setter Encapsulation k parts hote hain jo private data ko access or modify krne k lia use hote hain

i). **Getter** private variable ki value read or get krne k lia use hota hai

ii). **Setter** private variable ki value change ya update krne k lia use hota hai

ii. Inheritance aik oop concept hai jis may child class parent class k

methods or properties ko inherit krte hai

Type of inheritance

- **Single Inheritance**
- **Multi-level inheritance**
- **Multiple inheritance**
 - a) **Single-Inheritance** Jb aik child class aik parent class se inherit krte hai
 - b) **Multi-level Inheritance** jb aik child class se parent class se or parent class grand parent se inherit kre
 - c) **Multiple Inheritance** Jab multiple child classes ek hi parent class se inherit krte hain.